

DNAX™ ASSAY PROTOCOL & MOLECULAR SPECIFICATION REPORT

Construct Designation: DNAX_Validation_Construct_01 | Architecture: LINEAR | Generated: 2026-08-21 15:14:56 UTC

1. CONSTRUCT BIOPHYSICAL PROPERTIES

Construct ID	DNAX_Validation_Construct_01	Architecture	LINEAR
Payload Length	500 bp	Molecular Weight (MW)	330.00 kDa
GC Content	51.4%	Copy Number / ng	1.82e+09 copies/ng
In Silico Homology	Passed (≥ 25% Unique)	Local DB Clashes	0 Matches (Orthogonal)

2. PCR AMPLIFICATION PRIMER PAIRS

Oligo	Sequence (5' → 3')	Length	Tm (°C)	GC%	Target Strand
Forward Primer	CGATCGATCGATCGATCGAT	20 bp	59.2°C	50.0%	Sense (5' → 3')
Reverse Primer	TAACGATCGATCGCTAGCGC	20 bp	58.8°C	50.0%	Antisense (3' ← 5')

Amplicon Size: 500 bp | Primer Tm Matching Delta (ΔTm): 0.40°C (Optimal ≤ 1.5°C)

3. 4-CHANNEL MULTIPLEX TAQMAN PROBE SET

Channel	Reporter / Quencher	Probe Sequence (5' → 3')	Length	Tm (°C)	GC%	Coordinates
FAM	FAM / BHQ-1	CATGCGATCGATCGATCGATCGAT	24 bp	69.5°C	50.0%	bp 30–54
HEX	HEX / BHQ-1	AGCTAGCTAGCTAGCTAGCTAGCT	24 bp	70.1°C	48.0%	bp 80–104
ROX	ROX / BHQ-2	CGATCGATCGATCGATCGATCGAT	24 bp	69.8°C	52.0%	bp 140–164
Cy5	Cy5 / BHQ-3	TGCATGCATGCATGCATGCATGCA	24 bp	70.4°C	50.0%	bp 200–224

4. qPCR REACTION MASTER MIX RECIPE (20 µL REACTION)

Reagent Component	Stock Conc.	Final Conc.	Vol / 1 Rxn (µL)	Vol / 10 Rxns (µL)
2X TaqMan Fast Advanced Master Mix	2X	1X	10.0 µL	100.0 µL
Forward Primer (5'-3')	10 µM	400 nM	0.8 µL	8.0 µL
Reverse Primer (5'-3')	10 µM	400 nM	0.8 µL	8.0 µL
Multiplex TaqMan Probe (Selected Channel)	10 µM	200 nM	0.4 µL	4.0 µL
Synthetic DNA Template	10^4 copies/µL	10^3–10^5 copies	2.0 µL	20.0 µL
Nuclease-Free ddH2O	--	--	6.0 µL	60.0 µL
TOTAL REACTION VOLUME	--	--	20.0 µL	200.0 µL

5. OPTIMIZED THERMOCYCLING PROGRAM (REAL-TIME PCR)

Stage / Step	Temperature (°C)	Time Duration	Cycles	Optical Data Acquisition
1. UDG Decontamination	50.0°C	2 minutes	1 cycle	Disabled
2. DNA Polymerase Activation	95.0°C	20 seconds	1 cycle	Disabled (Hot-Start)
3. Denaturation	95.0°C	3 seconds	40 cycles	Disabled
4. Anneal / Extension	58.0°C	30 seconds	40 cycles	Acquire (FAM, HEX, ROX, Cy5)

6. COMPLETE NUCLEOTIDE SEQUENCE MANIFEST (5' → 3')

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1 AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA
51 AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA
101 AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA
151 AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA
201 AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA
251 AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA
301 AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA
351 AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA
401 AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA
451 AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA AAAAAAAAAA
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